

Trial design and patients

Patients were enrolled at 6 sites: Minneapolis Heart Institute, Cedars-Sinai Medical Center, Mercy Gilbert Medical Center, University of Utah, University of Florida, and Rush University Medical Center.

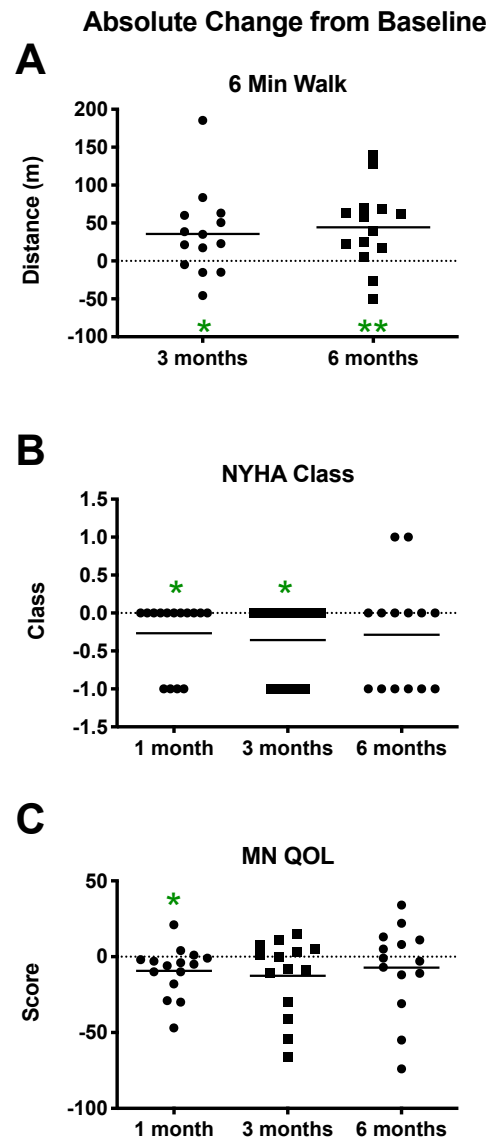
Inclusion Criteria:

1. The subject is 30-75 years of age
2. The subject must be able to provide informed consent
3. At least 60 days and no more than 3 years will have passed since the first ST elevation myocardial infarction (Index STEMI) at time of VentriGel administration
4. The Index STEMI must meet the following criteria:
 1. First time diagnosis of STEMI AND;
 2. Meet the STEMI criteria of the American College of Cardiology (ACC)/American Heart Association (AHA) (e.g. ST elevation in at least 2 contiguous leads >0.2 mV in V1, V2 or V3 and/or >0.1 mV in at least two other leads), or new left bundle branch block (LBBB)
5. Evidence of left ventricular remodeling secondary to the myocardial infarction using 2-D echocardiography or cMR;
 1. the LVEF must be $\geq 25\%$ and $\leq 45\%$ AND;
 2. The left ventricular wall thickness is ≥ 8 mm in target area.
6. Successful percutaneous coronary intervention (PCI) restoring TIMI II or higher flow to infarcted area
7. Negative pregnancy test [serum human chorionic gonadotropin (β hCG)] in women of childbearing potential within 24 hours prior to dosing) or if less than 2 years postmenopausal agree to use of adequate contraception during the study.
8. Must be ambulatory, willing and able to comply with protocol, including follow-up visits
9. Subject must be receiving best medical treatment for their post-MI clinical presentation according to the American College of Cardiology (ACC)/American Heart Association (AHA) guidelines
10. For those subjects indicated for heart failure medical therapy, subjects must be on stable therapy including beta-blockers and angiotensin converting enzyme inhibitors, if tolerated, for at least 45 days prior to therapy delivery

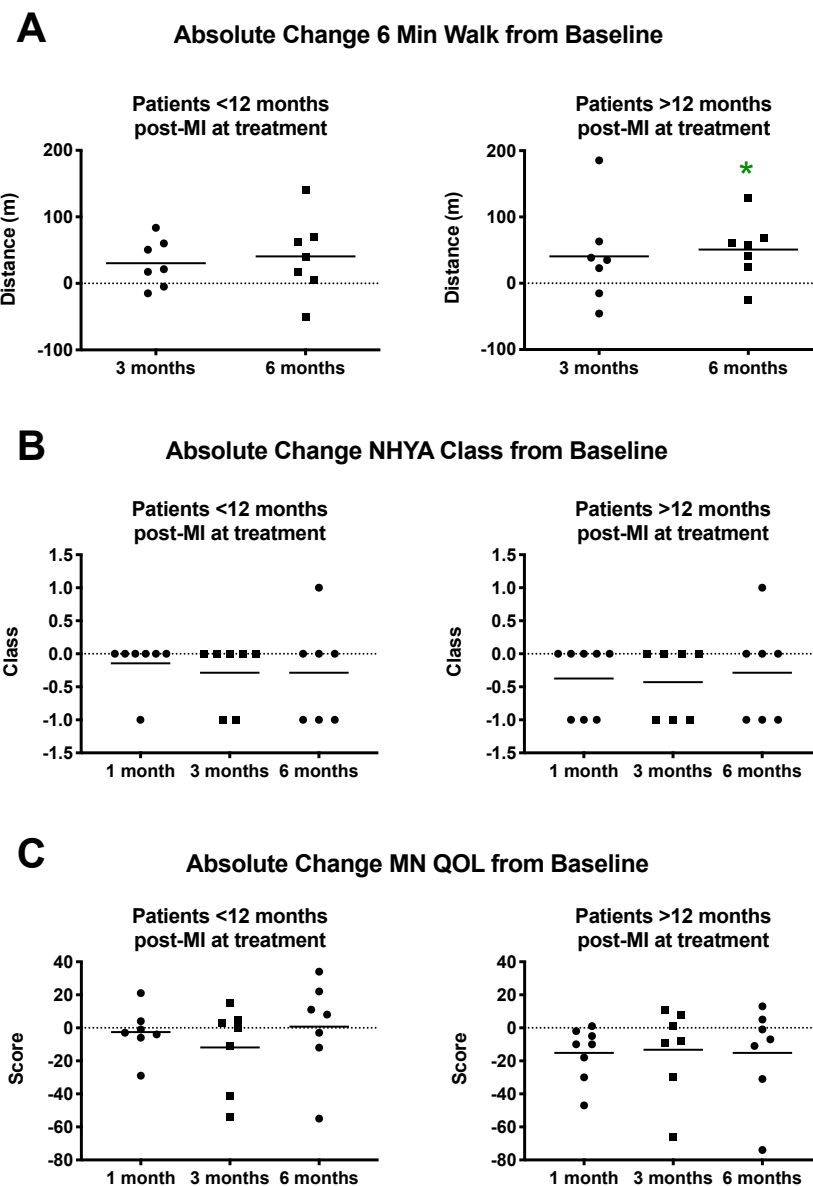
Exclusion Criteria:

1. Contraindications to cardiac MR
2. NYHA Functional Classification 4 heart failure within the prior 6 months.
3. Significant coronary artery stenosis that may require percutaneous or surgical revascularization within six months of enrollment, as determined by the principal investigator
4. Left ventricular thrombus, left ventricular aneurysm, subjects with post-infarction pericarditis, or subjects with wall motion abnormalities outside the region of the infarct related artery
5. Frequent, recurrent, sustained (>30 seconds) ventricular tachycardia in 30 days prior to VentriGel administration
6. ECG or 24 hour Holter Monitor with any of the following findings:

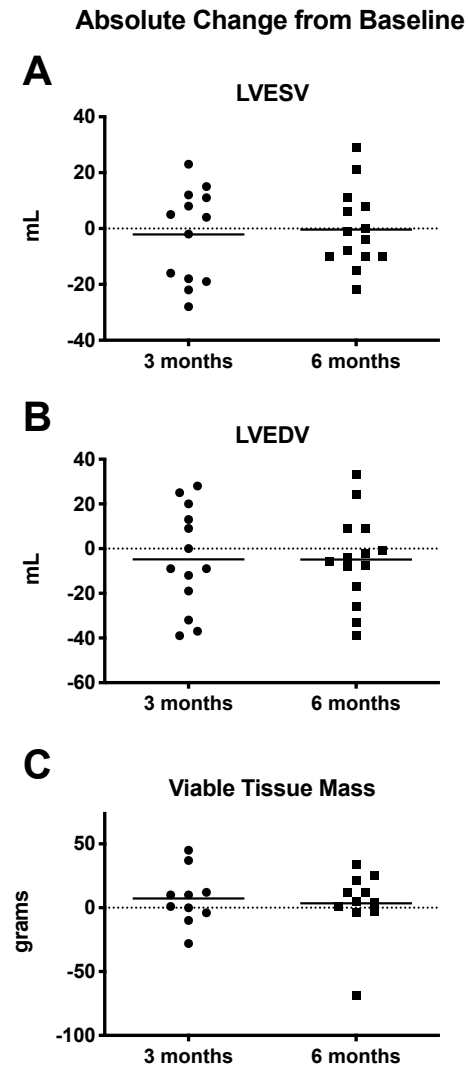
1. Bifascicular block (left bundle branch block or right bundle branch block plus left hemi-block)
2. Higher grade AV block (i.e. 3rd degree)
3. Ventricular tachycardia (≥ 5 seconds of VT OR any symptomatic VT)
7. Atrial fibrillation with heart rate greater than 110 bpm.
8. Severe valvular disease (e.g. aortic stenosis of moderate or worse severity, valvular insufficiency requiring surgical repair) or history of heart valve replacement.
9. Known allergy to porcine proteins or prior implantation of a porcine derived medical product including cardiac valves or other ECM products.
10. Etiology of heart failure due to any cause (e.g. hypertrophic cardiomyopathies, restrictive cardiomyopathies, constrictive pericardial disease, amyloidosis, active myocarditis) other than the index MI.
11. Severe peripheral vascular disease that impairs femoral arterial access.
12. Less than 3 years, cancer free, since end of treatment for cancer (with exception of basal cell carcinoma)
13. Alcohol or drug dependency within six months prior to enrollment
14. Cerebrovascular event within the 90 days prior or major surgical procedure or major trauma within the 14 days prior to enrollment
15. Participation, defined as receiving test article, in an experimental clinical study within 30 days prior to administration of VentriGel (i.e. screen failure from other study does not exclude subject)
16. Uncontrolled hypertension defined as systolic blood pressure (SBP) > 180 mmHg and/or or diastolic blood pressure (DBP) > 110 mmHg
17. Abnormal laboratory values as defined below performed at screening:
 1. Aspartate aminotransferase [AST]/ alanine aminotransferase [ALT] ≥ 3 times upper limit of normal (ULN)
 2. Serum creatinine ≥ 2.0 mg/dL
 3. Platelet count $< 50,000/mm^3$
 4. Hemoglobin < 9.0 g/dL
 5. HbA1c $> 9.0\%$
 6. PT or aPTT with clinically significant elevations relative to local laboratory norms
18. Any other cardiac or non-cardiac conditions or illness which, in the opinion of the principal investigator, may place subjects at undue risk or compromise the objectives of the study.
19. Institutional interpretation of cMR EF data outside the $\geq 25\%$ and $\leq 45\%$ limits



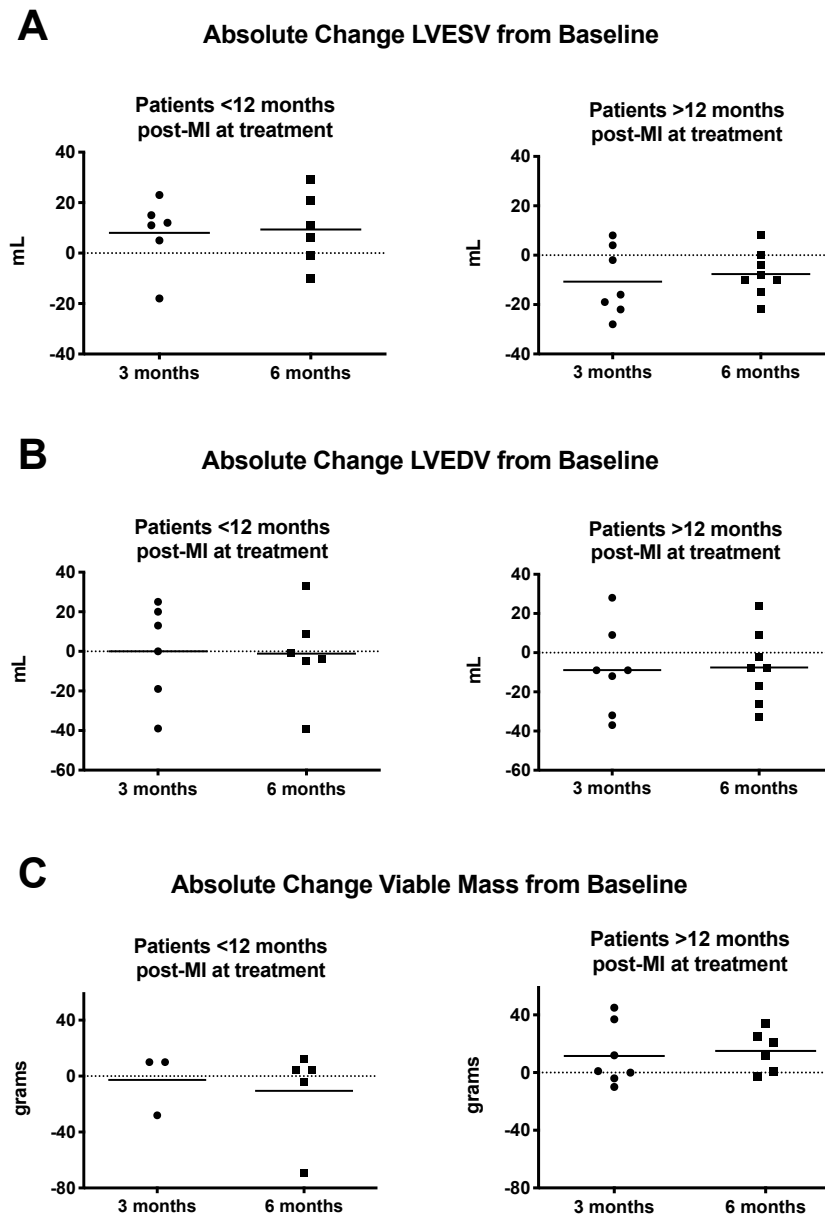
Online Figure 1. Scatter plots for changes in exercise capacity, heart failure class and quality of life. Scatter plots for absolute changes in (A) 6-min walk test distance, (B) New York Heart Association (NYHA) class, and (C) Minnesota Living with Heart Failure Questionnaire (MNLWHFQ) score from baseline to follow-up visits are shown for the total cohort. The solid horizontal line represents the mean. * $P < 0.05$ and ** $P < 0.01$ for paired t-test between follow-up and baseline.



Online Figure 2. Scatter plots for changes in exercise capacity, heart failure class and quality of life in early and late MI patients. Scatter plots for absolute changes in (A) 6-min walk test distance, (B) New York Heart Association (NYHA) class, and (C) Minnesota Living with Heart Failure Questionnaire (MNLWHFQ) score from baseline to follow-up visits are shown for patients <12 months post-MI and >12 months post-MI at treatment. The solid horizontal line represents the mean. * $P < 0.05$ for paired t-test between follow-up and baseline.



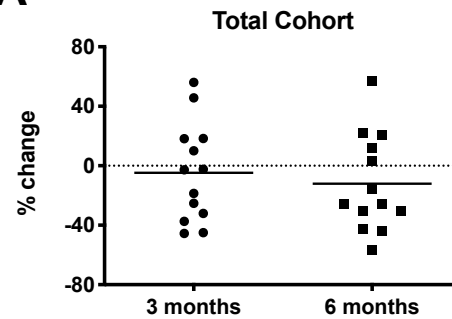
Online Figure 3. Scatter plots for magnetic resonance imaging parameters. Scatter plots for absolute changes in (A) left ventricular end-systolic volume (LVESV), (B) left ventricular end-systolic volume (LVEDV), and (C) viable mass from baseline to follow-up visits are shown for the total cohort. The solid horizontal line represents the mean.



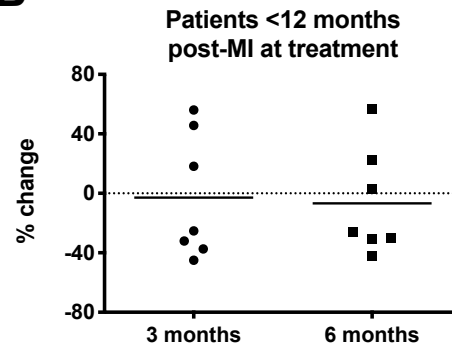
Online Figure 4. Scatter plots for magnetic resonance imaging parameters in early and late MI patients. Scatter plots for absolute changes in (A) left ventricular end-systolic volume (LVESV), (B) left ventricular end-systolic volume (LVEDV), and (C) viable mass from baseline to follow-up visits are shown for patients <12 months post-MI and >12 months post-MI at treatment. The solid horizontal line represents the mean.

Percent Change BNP from Baseline

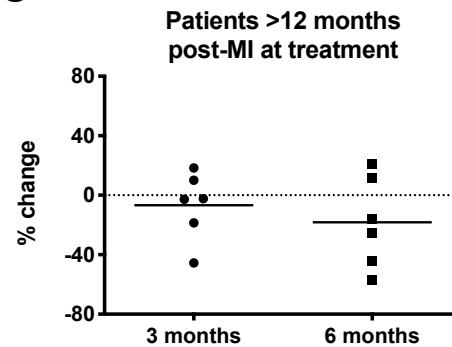
A



B



C



Online Figure 5. Scatter plots for B-type natriuretic peptide. Scatter plots for percent changes in B-type natriuretic peptide (BNP) from baseline to follow-up visits are shown for the (A) total cohort, and patients (B) <12 months and (C) >12 months post-MI at treatment. The solid horizontal line represents the mean.

Online Table 1 Patient Consent Dates

4-Sep-15
26-Aug-15
3-Aug-16
2-Nov-16
20-Dec-16
20-Mar-17
20-Apr-17
2-May-17
6-Jun-17
12-Jul-17
17-Jul-17
13-Jul-17
27-Jul-17
27-Jul-17
27-Jul-17

Online Table 2 Concomitant Medications

	All Patients	Patients <12 mo post-MI at treatment	Patients >12 mo post-MI at treatment	Drug elimination or reduction Patients <12 mo post-MI at treatment	Drug elimination or reduction Patients >12 mo post-MI at treatment	Drug addition or increase Patients <12 mo post-MI at treatment	Drug addition or increase Patients >12 mo post-MI at treatment
Patients with at least one medication	15 (100)	7 (100)	8 (100)	4 (57.1)	3 (37.5)	6 (85.7)	3 (37.5)
ACE inhibitors and calcium channel blockers	1 (6.7)	0 (0)	1 (12.5)	0 (0)	0 (0)	0 (0)	0 (0)
Amlodipine w/benazepril hydrochloride	1 (6.7)	0 (0)	1 (12.5)	0 (0)	0 (0)	0 (0)	0 (0)
ACE inhibitors, plain	13 (86.7)	5 (71.4)	7 (87.5)	0 (0)	0 (0)	0 (0)	0 (0)
Benazepril hydrochloride	1 (6.7)	1 (14.3)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Enalapril	1 (6.7)	0 (0)	1 (12.5)	0 (0)	0 (0)	0 (0)	0 (0)
Lisinopril	9 (60.0)	4 (57.1)	5 (62.5)	0 (0)	0 (0)	0 (0)	0 (0)
Ramipril	2 (13.3)	0 (0)	2 (25.0)	0 (0)	0 (0)	0 (0)	0 (0)
Adrenergics in combination with anticholinergics	1 (6.7)	0 (0)	1 (12.5)	0 (0)	0 (0)	0 (0)	1 (12.5)
Umeclidinium bromide w/vilanterol trifenate	1 (6.7)	0 (0)	1 (12.5)	0 (0)	0 (0)	0 (0)	1 (12.5)
Aldosterone antagonists	6 (40.0)	1 (14.3)	5 (62.5)	0 (0)	0 (0)	0 (0)	0 (0)
Eplerenone	1 (6.7)	1 (14.3)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Spironolactone	5 (33.3)	0 (0)	5 (62.5)	0 (0)	0 (0)	0 (0)	0 (0)
Alpha and beta blocking agents	11 (73.3)	4 (57.1)	7 (87.5)	0 (0)	1 (12.5)	1 (14.3)	0 (0)
Carvedilol	11 (73.3)	4 (57.1)	7 (87.5)	0 (0)	1 (12.5)	1 (14.3)	0 (0)
Angiotensin ii antagonists, plain	1 (6.7)	1 (14.3)	0 (0)	1 (14.3)	0 (0)	1 (14.3)	0 (0)
Losartan	1 (6.7)	1 (14.3)	0 (0)	1 (14.3)	0 (0)	0 (0)	0 (0)
Valsartan	1 (6.7)	1 (14.3)	0 (0)	0 (0)	0 (0)	1 (14.3)	0 (0)
Anilides	1 (6.7)	0 (0)	1 (12.5)	0 (0)	0 (0)	0 (0)	0 (0)
Paracetamol	1 (6.7)	0 (0)	1 (12.5)	0 (0)	0 (0)	0 (0)	0 (0)
Antiallergic agents, excl. corticosteroids	1 (6.7)	0 (0)	1 (12.5)	0 (0)	0 (0)	0 (0)	0 (0)
Azelastine hydrochloride	1 (6.7)	0 (0)	1 (12.5)	0 (0)	0 (0)	0 (0)	0 (0)
Antiarrhythmics, class Ib	1 (6.7)	1 (14.3)	0 (0)	0 (0)	0 (0)	1 (14.3)	0 (0)
Mexiletine	1 (6.7)	1 (14.3)	0 (0)	0 (0)	0 (0)	1 (14.3)	0 (0)
Anticholinergics	1 (6.7)	0 (0)	1 (12.5)	0 (0)	1 (12.5)	0 (0)	0 (0)
Tiotropium bromide	1 (6.7)	0 (0)	1 (12.5)	0 (0)	1 (12.5)	0 (0)	0 (0)
Ascorbic acid (vitamin c), combinations	2 (13.3)	2 (28.6)	0 (0)	0 (0)	0 (0)	1 (14.3)	0 (0)
Ascorbic acid (vitamin c), combinations	2 (13.3)	2 (28.6)	0 (0)	0 (0)	0 (0)	1 (14.3)	0 (0)

Ascorbic acid (vitamin c), plain	1 (6.7)	0 (0)	1 (12.5)	0 (0)	0 (0)	0 (0)	0 (0)
Ascorbic acid	1 (6.7)	0 (0)	1 (12.5)	0 (0)	0 (0)	0 (0)	0 (0)
Beta blocking agents, selective	6 (40.0)	4 (57.1)	2 (25.0)				
Metoprolol	3 (20.0)	2 (28.6)	1 (12.5)				
Metoprolol succinate	3 (20.0)	2 (28.6)	1 (12.5)	0 (0)	0 (0)	0 (0)	0 (0)
Biguanides	2 (13.3)	2 (28.6)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Metformin	2 (13.3)	2 (28.6)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Bulk-forming laxatives	1 (6.7)	1 (14.3)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Carmellose	1 (6.7)	1 (14.3)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Calcium	2 (13.3)	1 (14.3)	1 (12.5)	0 (0)	0 (0)	0 (0)	0 (0)
Calcium	2 (13.3)	1 (14.3)	1 (12.5)	0 (0)	0 (0)	0 (0)	0 (0)
Combinations of sulfonamides and trimethoprim, incl. Derivatives	1 (6.7)	1 (14.3)	0 (0)	0 (0)	0 (0)	1 (14.3)	0 (0)
Bactrim	1 (6.7)	1 (14.3)	0 (0)	0 (0)	0 (0)	1 (14.3)	0 (0)
Contact laxatives	2 (13.3)	1 (14.3)	1 (12.5)	0 (0)	0 (0)	0 (0)	0 (0)
Sennoside a+b	1 (6.7)	1 (14.3)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Senokot comfort	1 (6.7)	0 (0)	1 (12.5)	0 (0)	0 (0)	0 (0)	0 (0)
Corticosteroids	1 (6.7)	1 (14.3)	0 (0)	1 (14.3)	0 (0)	0 (0)	0 (0)
Duonase	1 (6.7)	1 (14.3)	0 (0)	1 (14.3)	0 (0)	0 (0)	0 (0)
Corticosteroids, moderately potent (group ii)	1 (6.7)	0 (0)	1 (12.5)	0 (0)	0 (0)	0 (0)	0 (0)
Triamcinolone acetonide	1 (6.7)	0 (0)	1 (12.5)	0 (0)	0 (0)	0 (0)	0 (0)
Dihydropyridine derivatives	1 (6.7)	0 (0)	1 (12.5)	0 (0)	0 (0)	0 (0)	0 (0)
Amlodipine	1 (6.7)	0 (0)	1 (12.5)	0 (0)	0 (0)	0 (0)	0 (0)
Direct factor xa inhibitors	2 (13.3)	1 (14.3)	1 (12.5)	0 (0)	0 (0)	0 (0)	1 (12.5)
Apixaban	2 (13.3)	1 (14.3)	1 (12.5)	0 (0)	0 (0)	0 (0)	1 (12.5)
Drugs used in erectile dysfunction	1 (6.7)	0 (0)	1 (12.5)	0 (0)	0 (0)	0 (0)	0 (0)
Alprostadil	1 (6.7)	0 (0)	1 (12.5)	0 (0)	0 (0)	0 (0)	0 (0)
Sildenafil citrate	1 (6.7)	0 (0)	1 (12.5)	0 (0)	0 (0)	0 (0)	0 (0)
Glucocorticoids	4 (26.7)	2 (28.6)	2 (25.0)	0 (0)	0 (0)	2 (28.6)	0 (0)
Fluticasone propionate	1 (6.7)	0 (0)	1 (12.5)	0 (0)	0 (0)	0 (0)	0 (0)
Mometasone furoate	1 (6.7)	0 (0)	1 (12.5)	0 (0)	0 (0)	0 (0)	0 (0)
Prednisone	2 (13.3)	2 (28.6)	0 (0)	0 (0)	0 (0)	2 (28.6)	0 (0)
H2-receptor antagonists	2 (13.3)	1 (14.3)	1 (12.5)	0 (0)	0 (0)	0 (0)	0 (0)
Ranitidine	2 (13.3)	1 (14.3)	1 (12.5)	0 (0)	0 (0)	0 (0)	0 (0)

Hmg coa reductase inhibitors	13 (86.7)	6 (85.7)	7 (87.5)	0 (0)	0 (0)	0 (0)	0 (0)
Atorvastatin	12 (80.0)	6 (85.7)	6 (75.0)	0 (0)	0 (0)	0 (0)	0 (0)
Rosuvastatin	1 (6.7)	0 (0)	1 (12.5)	0 (0)	0 (0)	0 (0)	0 (0)
Iron bivalent, oral preparations	1 (6.7)	0 (0)	1 (12.5)	0 (0)	0 (0)	0 (0)	0 (0)
Ferrous sulfate	1 (6.7)	0 (0)	1 (12.5)	0 (0)	0 (0)	0 (0)	0 (0)
Leukotriene receptor antagonists	1 (6.7)	0 (0)	1 (12.5)	0 (0)	0 (0)	0 (0)	0 (0)
Montelukast	1 (6.7)	0 (0)	1 (12.5)	0 (0)	0 (0)	0 (0)	0 (0)
Magnesium	2 (13.3)	2 (28.6)	0 (0)	0 (0)	0 (0)	2 (28.6)	0 (0)
Magnesium	2 (13.3)	2 (28.6)	0 (0)	0 (0)	0 (0)	2 (28.6)	0 (0)
Organic nitrates	8 (53.3)	4 (57.1)	4 (50.0)	0 (0)	0 (0)	0 (0)	0 (0)
Glyceryl trinitrate	8 (53.3)	4 (57.1)	4 (50.0)	0 (0)	0 (0)	0 (0)	0 (0)
Other antiinflammatory and antirheumatic agents, non-steroids	2 (13.3)	1 (14.3)	1 (12.5)	0 (0)	0 (0)	0 (0)	0 (0)
Chondroitin sulfate w/glucosamine	2 (13.3)	1 (14.3)	1 (12.5)	0 (0)	0 (0)	0 (0)	0 (0)
Other cardiac preparations	1 (6.7)	0 (0)	1 (12.5)	0 (0)	0 (0)	0 (0)	0 (0)
Ivabradine hydrochloride	1 (6.7)	0 (0)	1 (12.5)	0 (0)	0 (0)	0 (0)	0 (0)
Other lipid modifying agents	5 (33.3)	2 (28.6)	3 (37.5)	0 (0)	1 (12.5)	1 (14.3)	0 (0)
Evolocumab	1 (6.7)	1 (14.3)	0 (0)	0 (0)	0 (0)	1 (14.3)	0 (0)
Ezetimibe	1 (6.7)	0 (0)	1 (12.5)	0 (0)	0 (0)	0 (0)	0 (0)
Fish oil	3 (20.0)	1 (14.3)	2 (25.0)	0 (0)	1 (12.5)	0 (0)	0 (0)
Other opioids	1 (6.7)	1 (14.3)	0 (0)	0 (0)	0 (0)	1 (14.3)	0 (0)
Tramadol	1 (6.7)	1 (14.3)	0 (0)	0 (0)	0 (0)	1 (14.3)	0 (0)
Piperazine derivatives	1 (6.7)	0 (0)	1 (12.5)	0 (0)	0 (0)	0 (0)	0 (0)
Cetirizine	1 (6.7)	0 (0)	1 (12.5)	0 (0)	0 (0)	0 (0)	0 (0)
Platelet aggregation inhibitors excl. heparin	15 (100)	7 (100)	8 (100)	1 (14.3)	1 (12.5)	2 (28.6)	0 (0)
Acetylsalicylic acid	15 (100)	7 (100)	8 (100)	0 (0)	0 (0)	2 (28.6)	0 (0)
Clopidogrel	6 (40.0)	2 (28.6)	4 (50.0)	0 (0)	1 (12.5)	0 (0)	0 (0)
Clopidogrel bisulfate	2 (13.3)	2 (28.6)	0 (0)	1 (14.3)	0 (0)	0 (0)	0 (0)
Prasugrel hydrochloride	1 (6.7)	1 (14.3)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Ticagrelor	3 (20.0)	2 (28.6)	1 (12.5)	0 (0)	0 (0)	0 (0)	0 (0)
Potassium	3 (20.0)	2 (28.6)	1 (12.5)	1 (14.3)	0 (0)	1 (14.3)	0 (0)
Potassium chloride	3 (20.0)	2 (28.6)	1 (12.5)	1 (14.3)	0 (0)	1 (14.3)	0 (0)
Proton pump inhibitors	2 (13.3)	1 (14.3)	0 (0)	0 (0)	0 (0)	1 (14.3)	0 (0)
Omeprazole	1 (6.7)	1 (14.3)	0 (0)	0 (0)	0 (0)	1 (14.3)	0 (0)
Pantoprazole	1 (6.7)	0 (0)	1 (12.5)	0 (0)	0 (0)	0 (0)	0 (0)

Selective beta-2-adrenoreceptor agonists	2 (13.3)	0 (0)	2 (25.0)	0 (0)	0 (0)	0 (0)	0 (0)
Salbutamol	2 (13.3)	0 (0)	2 (25.0)	0 (0)	0 (0)	0 (0)	0 (0)
Softeners, emollients	1 (6.7)	0 (0)	1 (12.5)	0 (0)	0 (0)	0 (0)	0 (0)
Docusate sodium	1 (6.7)	0 (0)	1 (12.5)	0 (0)	0 (0)	0 (0)	0 (0)
Sulfonamides, plain	3 (20.0)	1 (14.3)	2 (25.0)	0 (0)	0 (0)	0 (0)	1 (12.5)
Furosemide	3 (20.0)	1 (14.3)	2 (25.0)	0 (0)	0 (0)	0 (0)	1 (12.5)
Furosemide sodium	1 (6.7)	1 (14.3)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Toraseamide	1 (6.7)	0 (0)	1 (12.5)	0 (0)	0 (0)	0 (0)	0 (0)
Sulfonylureas	1 (6.7)	1 (14.3)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Glipizide	1 (6.7)	1 (14.3)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Tetracyclines	1 (6.7)	0 (0)	1 (12.5)	0 (0)	0 (0)	0 (0)	0 (0)
Minocycline hydrochloride	1 (6.7)	0 (0)	1 (12.5)	0 (0)	0 (0)	0 (0)	0 (0)
Thyroid hormones	1 (6.7)	1 (14.3)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Levothyroxine sodium	1 (6.7)	1 (14.3)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Vitamin b-complex, other combinations	2 (13.3)	0 (0)	2 (25.0)	0 (0)	0 (0)	0 (0)	0 (0)
Multivitamin	2 (13.3)	0 (0)	2 (25.0)	0 (0)	0 (0)	0 (0)	0 (0)
Vitamin b-complex, plain	1 (6.7)	0 (0)	1 (12.5)	0 (0)	0 (0)	0 (0)	0 (0)
Vitamin b complex	1 (6.7)	0 (0)	1 (12.5)	0 (0)	0 (0)	0 (0)	0 (0)
Vitamin d and analogues	3 (20.0)	2 (28.6)	1 (12.5)	0 (0)	0 (0)	1 (14.3)	0 (0)
Colecalciferol	2 (13.3)	1 (14.3)	1 (12.5)	0 (0)	0 (0)	0 (0)	0 (0)
Vitamin d nos	1 (6.7)	1 (14.3)	0 (0)	0 (0)	0 (0)	1 (14.3)	0 (0)
Vitamin k antagonists	2 (13.3)	2 (28.6)	0 (0)	1 (14.3)	0 (0)	1 (14.3)	0 (0)
Warfarin	2 (13.3)	2 (28.6)	0 (0)	1 (14.3)	0 (0)	1 (14.3)	0 (0)

Values are n (%). % in Drug elimination/addition columns is % of total patients in each subset (either <12 mo post-MI or >12 mo post MI).

Online Table 3 Adverse Events

	All Patients (N=15)	Relationship to VentriGel
Number of patients with at least one adverse event	11 (73.3)	
Blood and lymphatic system disorders	1 (6.7)	Not related
Thrombocytopenia	1 (6.7)	
Cardiac disorders	6 (40.0)	
Cardiogenic shock	1 (6.7)	Possible
Complete heart block	1 (6.7)	Possible
Heart failure	1 (6.7)	Not related
Heart failure NYHA class IV	1 (6.7)	Not related
Intracardiac thrombus	2 (13.3)	Not related
Non-sustained ventricular tachycardia	1 (6.7)	Not related
Paroxysmal atrial fibrillation	1 (6.7)	Not related
Gastrointestinal disorders	3 (20.0)	Not related
Abdominal distension	1 (6.7)	
Acid reflux (esophageal)	1 (6.7)	
Diarrhea	1 (6.7)	
General disorders and administration site conditions	2 (13.3)	Not related
Edema aggravated	1 (6.7)	
Stent occlusion	1 (6.7)	
Hepatobiliary disorders	1 (6.7)	Not related
Perforation of gallbladder	1 (6.7)	
Infections and infestations	2 (13.3)	Not related
Head cold	1 (6.7)	
Urinary tract infection	1 (6.7)	
Investigations	2 (13.3)	Not related
Platelet count decreased	1 (6.7)	
Systolic murmur	1 (6.7)	
Metabolism and nutrition disorders	1 (6.7)	Not related
Hyperglycemia	1 (6.7)	
Musculoskeletal and connective tissue disorders	1 (6.7)	Not related
Muscle spasms	1 (6.7)	
Pain in hip	1 (6.7)	

Neoplasms benign, malignant and unspecified (incl cysts and polyps)	1 (6.7)	
Colon cancer	1 (6.7)	Unlikely
Nervous system disorders	3 (20.0)	Not related
Dizziness	1 (6.7)	
Light headedness	1 (6.7)	
Sciatica	1 (6.7)	
Psychiatric disorders	1 (6.7)	Not related
Delirium	1 (6.7)	
Respiratory, thoracic and mediastinal disorders	4 (26.7)	
Dyspnea exacerbated	1 (6.7)	Not related
Egophony	1 (6.7)	Not related
Increased shortness of breath	1 (6.7)	Not related
Productive cough	1 (6.7)	Unlikely
Rhinorrhea	1 (6.7)	Unlikely
Shortness of breath	1 (6.7)	Not related
Skin and subcutaneous tissue disorders	2 (13.3)	Not related
Ecchymosis	1 (6.7)	
Skin rash	1 (6.7)	
Surgical and medical procedures	1 (6.7)	Not related
Implantable cardioverter defibrillator insertion	1 (6.7)	

Values are n (%). NYHA = New York Heart Association

Online Table 4. C-Reactive Protein

	Baseline	1 day follow-up	Change from baseline to 1 day	Paired t-test <i>P</i> -value (1 day vs. baseline)	1 week follow-up	Change from baseline to 1 week	Paired t-test <i>P</i> -value (1 week vs. baseline)	2 week follow-up	Change from baseline to 2 week	Paired t-test <i>P</i> -value (2 week vs. baseline)
Total cohort	0.7 (0.4) n=11	2.6 (1.0) n=7	2.2 (1.1)	0.13	1.3 (0.7) n=10	0.9 (0.2)	0.73	0.9 (1.6) n=12	0.1 (0.2)	0.64
Patients <12 mo post-MI	1.3 (0.8) n=5	4.1 (1.9) n=3	3.3 (2.9)	0.47	2.0 (1.1) n=6	0.2 (0.4)	0.70	1.5 (0.9) n=6	0.2 (0.3)	0.54
Patients >12 mo post-MI	0.2 (0.1) n=6	1.5 (0.7) n=4	1.4 (0.9)	0.26	0.2 (0.1) n=4	-0.1 (0.0)	0.26	0.2 (0.1) n=6	-0.1 (0.0)	0.03

	1 month follow-up	Change from baseline to 1 month	Paired t-test <i>P</i> -value (1 month vs. baseline)	3 month follow-up	Change from baseline to 3 month	Paired t-test <i>P</i> -value (3 month vs. baseline)	6 month follow-up	Change from baseline to 6 month	Paired t-test <i>P</i> -value (6 month vs. baseline)	Repeated measures mixed effects model <i>P</i> -value
Total cohort	1.2 (0.6) n=12	0.4 (0.3)	0.25	1.0 (0.5) n=12	0.3 (0.2)	0.22	1.5 (0.6) n=12	-0.7 (0.4)	0.10	0.04
Patients <12 mo post-MI	2.1 (1.2) n=6	0.9 (0.7)	0.27	1.6 (0.9) n=6	0.3 (0.3)	0.41	2.2 (0.8) n=6	-1.3 (0.8)	0.06	0.09
Patients >12 mo post-MI	0.2 (0.1) n=6	0.0 (0.1)	0.92	0.4 (0.4) n=6	0.3 (0.4)	0.45	0.8 (0.8) n=6	-0.2 (0.1)	0.41	0.25

MI: myocardial infarction

Online Table 5. Functional exercise capacity

	Baseline	3 month follow-up	Change from baseline to 3 month	Paired t-test <i>P</i> -value (3 month vs. baseline)	6 month follow-up	Change from baseline to 6 month	Paired t-test <i>P</i> -value (6 month vs. baseline)	Repeated measures mixed effects model <i>P</i> -value
6 min walk test								
Total cohort	429.4 (24.8) n=15	465.9 (28.4) n=14	35.5 (14.9)	0.03	474.6 (30.3) n=14	44.4 (13.9)	0.007	0.004
Patients <12 mo post-MI	442.4 (28.9) n=7	472.9 (36.7) n=7	30.4 (13.5)	0.07	482.8 (43.5) n=7	40.5 (22.5)	0.12	0.11
Patients >12 mo post-MI	418.0 (40.5) n=8	458.9 (46.2) n=7	40.6 (27.8)	0.19	466.5 (45.4) n=7	50.9 (17.7)	0.04	0.13

MI: myocardial infarction

Online Table 6. NYHA class and Quality of Life

	Baseline	1 month follow-up	Change from baseline to 1 month	Paired t-test <i>P</i> -value (1 month vs. baseline)	3 month follow-up	Change from baseline to 3 month	Paired t-test <i>P</i> -value (3 month vs. baseline)	6 month follow-up	Change from baseline to 6 month	Paired t-test <i>P</i> -value (6 month vs. baseline)	Repeated measures mixed effects model <i>P</i> -value
NYHA Class											
Total cohort	1.9 (0.1) n=15	1.6 (0.2) n=15	-0.3 (0.1)	0.04	1.5 (0.2) n=14	-0.4 (0.1)	0.02	1.6 (0.3) n=14	-0.3 (0.2)	0.17	0.12
Patients <12 mo post-MI	2.0 (0.2) n=7	1.9 (0.3) n=7	-0.1 (0.1)	0.36	1.7 (0.3) n=7	-0.3 (0.2)	0.17	1.7 (0.4) n=7	-0.3 (0.3)	0.36	0.54
Patients >12 mo post-MI	1.8 (0.2) n=8	1.4 (0.2) n=8	-0.4 (0.2)	0.08	1.3 (0.2) n=7	-0.4 (0.2)	0.08	1.4 (0.3) n=7	-0.3 (0.3)	0.36	0.16
MN QOL											
Total cohort	38.9 (7.4) n=15	29.7 (7.5) n=15	-9.3 (4.2)	0.045	28.1 (7.4) n=14	-12.6 (6.8)	0.09	33.4 (8.4) n=14	-7.2 (7.8)	0.37	0.24
Patients <12 mo post-MI	44.3 (10.3) n=7	41.7 (12.8) n=7	-2.6 (5.6)	0.66	32.4 (11.3) n=7	-11.9 (9.8)	0.27	45.0 (13.3) n=7	0.7 (10.9)	0.95	0.54
Patients >12 mo post-MI	34.3 (11.0) n=8	19.1 (7.3) n=8	-15.1 (5.7)	0.03	23.7 (10.2) n=7	-13.3 (10.2)	0.24	21.9 (9.3) n=7	-15.1 (11.2)	0.22	0.21

NYHA: New York Heart Association; MN QOL: Minnesota Living with Heart Failure quality of life score; MI: myocardial infarction

Online Table 7. Magnetic resonance imaging results

	Baseline	3 month follow-up	Change from baseline to 3 month	Paired t-test <i>P</i> -value (3 month vs. baseline)	6 month follow-up	Change from baseline to 6 month	Paired t-test <i>P</i> -value (6 month vs. baseline)	Repeated measures mixed effects model <i>P</i> -value
LVESV (mL)								
Total cohort	148.5 (16.7) n=14	148.7 (17.9) n=13	-2.1 (4.6)	0.66	148.1 (17.5) n=14	-0.35 (3.8)	0.93	0.10
Patients <12 mo post-MI	156.8 (25.2) n=6	164.8 (27.1) n=6	8.0 (5.7)	0.22	166.2 (29.5) n=6	9.3 (5.8)	0.17	0.20
Patients >12 mo post-MI	142.3 (23.5) n=8	134.9 (24.3) n=7	-110.7 (5.3)	0.09	134.6 (21.6) n=8	-7.6 (3.2)	0.05	0.11
LVEDV (mL)								
Total cohort	229.6 (18.6) n=14	228.5 (19.1) n=13	-4.8 (6.4)	0.47	224.6 (19.0) n=14	-4.9 (5.4)	0.38	0.38
Patients <12 mo post-MI	245.3 (26.5) n=6	245.3 (28.0) n=6	0.0 (10.1)	>0.99	244.0 (30.1) n=6	-1.2 (9.5)	0.89	0.95
Patients >12 mo post-MI	217.8 (26.5) n=8	214.1 (26.9) n=7	-8.9 (8.5)	0.34	210.1 (24.4) n=8	-7.6 (6.5)	0.28	0.19
Ejection fraction (%)								
Total cohort	37.1 (2.3) n=14	37.0 (2.8) n=13	-0.33 (1.0)	0.77	35.8 (2.5) n=14	-1.3 (1.2)	0.29	0.34
Patients <12 mo post-MI	37.7 (3.6) n=6	34.5 (4.3) n=6	-3.2 (1.1)	0.03	33.8 (4.3) n=6	-3.8 (1.3)	0.03	0.02
Patients >12 mo post-MI	36.6 (3.3) n=8	39.1 (3.7) n=7	2.1 (1.0)	0.08	37.3 (3.1) n=8	-0.6 (1.5)	0.69	0.18
Viable mass (g)								
Total cohort	114.2 (6.8) n=12	125.7 (7.4) n=10	7.3 (6.8)	0.31	118.4 (6.3) n=12	3.4 (8.1)	0.68	0.48
Patients <12 mo post-MI	121.6 (14.8) n=5	138.0 (9.0) n=3	-2.7 (12.7)	0.85	111.2 (12.3) n=5	-10.5 (14.9)	0.52	0.54
Patients >12 mo post-MI	108.9 (5.6) n=7	120.4 (9.5) n=7	11.6 (8.0)	0.20	123.6 (6.4) n=7	15.0 (5.9)	0.05	0.14

Scar tissue (% LV)								
Total cohort	27.1 (1.8) n=12	26.1 (2.7) n=10	-0.80 (1.6)	0.55	28.0 (1.9) n=12	0.50 (1.9)	0.58	0.51
Patients <12 mo post-MI	25.9 (1.6) n=5	25.0 (2.0) n=3	0.5 (3.8)	0.91	28.2 (2.6) n=5	2.3 (3.5)	0.55	0.48
Patients >12 mo post-MI	27.9 (2.9) n=7	26.5 (3.8) n=7	-1.4 (1.9)	0.50	27.9 (2.8) n=7	-1.0 (1.9)	0.63	0.51

LVESV: left ventricular end-systolic volume; LVEDV: left ventricular end-diastolic volume; LV: left ventricle; MI: myocardial infarction

Online Table 8. B-type Natriuretic Peptide

	Baseline	1 day follow-up	% Change from baseline to 1 day	Paired t-test <i>P</i> -value (1 day vs. baseline)	1 week follow-up	% Change from baseline to 1 week	Paired t-test <i>P</i> -value (1 week vs. baseline)	1 month follow-up	% Change from baseline to 1 month	Paired t-test <i>P</i> -value (1 month vs. baseline)
Total cohort	294.8 (86.4) n=14	363.5 (108.0) n=13	11.1 (13.1)	0.38	419.6 (143.8) n=11	14.2 (16.7)	0.14	374.2 (119.0) n=15	28.1 (10.2)	0.07
Patients <12 mo post-MI	316.1 (152.6) n=7	384.2 (221.5) n=6	1.8 (14.1)	0.36	382.2 (235.3) n=6	-8.9 (10.2)	0.48	432.3 (240.7) n=7	21.2 (13.3)	0.24
Patients >12 mo post-MI	273.4 (94.2) n=7	305.1 (92.8) n=7	20.5 (22.9)	0.68	464.5 (172.5) n=5	41.9 (31.9)	0.24	323.4 (94.9) n=8	35.0 (16.1)	0.19

	3 month follow-up	% Change from baseline to 3 month	Paired t-test <i>P</i> -value (3 month vs. baseline)	6 month follow-up	% Change from baseline to 6 month	Paired t-test <i>P</i> -value (6 month vs. baseline)	Repeated measures mixed effects model <i>P</i> -value
Total cohort	328.9 (136.0) n=14	-4.7 (9.2)	0.42	312.0 (135.3) n=14	-12.0 (9.1)	0.63	0.22
Patients <12 mo post-MI	412.7 (252.5) n=7	-2.9 (15.9)	0.38	402.0 (253.0) n=7	-6.7 (13.6)	0.43	0.18
Patients >12 mo post-MI	245.0 (118.5) n=7	-6.8 (9.3)	0.82	222.0 (112.2) n=7	-18.3 (12.4)	0.41	0.32

MI: myocardial infarction